

Research Methods

BIT/BIT 3

Research in Information System/Technology Studies

- **Research**

- Systematic investigation into and study of materials and sources to *establish facts* and reach new conclusions.
- A study that aims to gather and analyse non-numerical (descriptive) data to gain *an understanding of individuals' social reality*, including understanding their attitudes, beliefs, and motivation.

- **Why research in IS/IT**

- Information systems are complex
 - User needs goes beyond functional requirements
- Technology transfer from the global north to global south requires proper understating of local context expectations
- Technology changes and user acceptance

Research in Information System/Technology Studies/2

- **Units of study in IT/IS research**

- an object of study within a research project. It is the smallest unit a researcher can use to identify and describe a phenomenon
- the 'what' or 'who' the researcher wants to study, specific examples include:
 - Users of a system
 - Managers
 - System
 - Practices
 - Processes

Research paradigms

- A research paradigm explains **what** you **believe** reality (knowledge) is and **how** you think knowledge can be understood.
- Two common terms /taxonomy relating to the belief
 - Ontology
 - Epistemology
- Ontology:
 - a system of belief that reflects an interpretation of an individual about what constitutes a knowledge. Fact or an understanding/meaning
- Epistemology:
 - the theory of knowledge that deals with how knowledge is gathered and from which sources. Measure, interpreted exists constructed

Research Approach

- The procedure selected by the researcher to collect, analyze, and interpret data.
 - The procedure is hugely tied to the researcher's ontology and epistemology
- There are three approaches to research:
 - Quantitative
 - Qualitative
 - Mixed methods:
- Which is the best approach
 - Battle of research fields
 - Not an issue of superiority
 - Organized way to get the result

Nature of qualitative research

- Qualitative research is a method to elicit humane experiences in their social context.
- Characteristics of qualitative research are:
 - flexibility of researcher and research process.
 - Inductive approach is the core of qualitative research.
 - Research process in qualitative research is iterative.
- Sociological approaches in qualitative research:
 - **Interpretivism** : sociological method of research in which an action or event is analyzed based on the beliefs, norms, and values of the culture of the society in which it takes place.
 - **Constructivism** : recognizing that knowledge is actively constructed by individuals through their experiences and interactions
- Holds the belief that knowledge is subjective

Nature of quantitative research

- Quantitative research deals in numbers, logic, and an objective stance.
- Characteristics of quantitative research are:
 - Researcher to tied to research process
 - Deductive approach is the core of quantitative research.
 - Research process in qualitative research sequential
- Holds the belief that knowledge exists
- Holds the belief that knowledge is objective

Nature of disciplined inquiry

- A discovery learning which is regulated by criteria for conducting research to create knowledge. Peirce called this the scientific method of resolving doubt to fixate belief.

What determines approach of study

- Problem, problem and problem
 - Determine the extent, the relationship, the cause
 - Explaining why, gain deep understanding, interpret people behaviour
- What you want to learn
 - Example in table

Quantitative	Qualitative
To what extent	Why do
How does X affect Y	How does
What caused X among	What are experiences

Assignment 1

- Identify a problem and suggest if this can be addressed through qualitative or quantitative research.
- Your report should have
 - One paragraph: not more than half a page explaining the problem
 - Questions (THREE) you may want to answer to solve the problem
 - The approach (nature of research) you would want to use with justification
- **Instruction**
 - No more than one page including your name and reg number.
 - Submitted online with Subject exactly : Assignement1_RM to mmsendema@mubas.ac.mw
 - Total Marks 25 Marks
 - Due date: 30th July 2024

Reference

- Open source: Use any reference available online